Machine Learning Classification of Glioblastoma Using Gene Expression Data

Author: Katarzyna Zielińska
Year: 2026

Introduction

Glioblastoma (GBM) is a highly aggressive and heterogeneous brain tumor characterized by substantial molecular and transcriptional differences between tumor and non-tumor tissues. Gene expression data can be used to identify molecular patterns associated with glioblastoma and to develop computational models capable of distinguishing tumor from non-tumor samples. The objective of this project was to develop and evaluate machine learning models for classification of glioblastoma samples based on gene expression data. The workflow included data preparation, probe-to-gene annotation, dimensionality reduction through feature selection, supervised classification using Logistic Regression and Random Forest, model evaluation, and stratified 5-fold cross-validation. The project demonstrates how machine learning methods can be applied to high-dimensional genomic data and how model performance can be evaluated using independent test data and cross-validation.

Biological Question

Can gene expression profiles be used to accurately distinguish glioblastoma samples from non-tumor brain tissue using machine learning methods?

Dataset

The project uses the publicly available GSE4290 dataset from the NCBI Gene Expression Omnibus (GEO).

Dataset information

- Database: NCBI Gene Expression Omnibus (GEO)
- Accession: GSE4290
- Organism: *Homo sapiens*
- Technology: Affymetrix Human Genome U133 Plus 2.0 Array
- Platform: GPL570

- Total samples analyzed: 100
- Glioblastoma samples: 77
- Non-tumor samples: 23

The original expression matrix contained 54,613 probes.
After probe annotation and aggregation of probes corresponding to the same gene, 22,189 genes were available for machine learning analysis.
The dataset was divided into two diagnostic groups:

Condition	Samples
Glioblastoma (GBM)	77
Non-tumor	23
Total	**100**

The original GEO data were downloaded directly from NCBI and are not included in the repository.

Methods

1. Data Preparation

The GSE4290 series matrix was loaded and sample metadata were extracted from the GEO dataset.
Samples were selected according to their histopathological diagnosis. Only two groups were retained:

- glioblastoma, grade 4
- non-tumor

Other diagnostic categories present in the original dataset were excluded from the classification analysis.
The final dataset contained 100 samples:

- 77 GBM
- 23 non-tumor

No missing expression values were detected in the selected dataset.

2. Probe Annotation

The original expression data were provided at the probe level.
The GPL570 Affymetrix annotation was used to map probe identifiers to gene symbols.
A total of approximately 45,000 probes could be mapped to gene identifiers.
When multiple probes corresponded to the same gene, their expression measurements were aggregated to obtain a gene-level expression matrix.
The resulting dataset contained:
22,189 genes × 100 samples

3. Feature Selection

The dataset contained more than 22,000 genes but only 100 biological samples. Directly training machine learning models using all genes would therefore create a very high-dimensional problem and could increase the risk of overfitting.

Feature selection was performed using:
SelectKBest + ANOVA F-test (f_classif)

The 100 genes with the highest F-scores were selected as machine learning features.
The feature selection step was incorporated into the machine learning pipeline so that feature selection was performed using the training data during model evaluation, reducing the risk of information leakage.
The top-ranked genes included, among others:

- RASAL1
- KCTD4
- TLN2
- NRXN3
- SST
- SEC61A2
- LRP11
- GSKIP
- LRRC8B
- WDR7

4. Train/Test Split

The dataset was divided into training and test sets using a **stratified 80/20 split**.
The training set contained:

- 80 samples
- 62 GBM
- 18 non-tumor

The test set contained:

- 20 samples
- 15 GBM
- 5 non-tumor

Stratification was used to preserve the proportion of the two diagnostic groups.

5. Machine Learning Models

Two supervised machine learning algorithms were trained.

Logistic Regression

Logistic Regression was used as a linear classification model. It provides a relatively simple and interpretable baseline for binary classification and is commonly used for high-dimensional biological datasets.

Random Forest

Random Forest was used as a nonlinear ensemble classification method. The algorithm combines multiple decision trees and can capture nonlinear relationships between gene expression features and the diagnostic class. Both models were implemented using **scikit-learn** pipelines together with the feature-selection step.

6. Model Evaluation

The models were evaluated using:

- Accuracy

- Precision
- Recall
- F1-score
- ROC-AUC

Confusion matrices were generated to visualize classification errors.
ROC curves were used to compare the ability of the models to distinguish GBM from non-tumor samples.

7. Cross-Validation

To obtain a more robust estimate of model performance, **stratified 5-fold cross-validation** was performed. The dataset was divided into five stratified subsets, with each subset being used once as the validation fold while the remaining four subsets were used for training. Mean performance and standard deviation were calculated for each metric. This analysis provides a more stable estimate of model performance than relying only on a single train/test split.

Software and Packages

The analysis was performed using:

- Python 3.11
- Linux / Ubuntu
- WSL2
- NumPy
- pandas
- scikit-learn
- Matplotlib
- Seaborn
- PyYAML
- joblib

Main tools

pandas - loading, processing and manipulation of gene expression and metadata tables.

NumPy - numerical operations and matrix processing.
scikit-learn - implementation of machine learning algorithms, feature selection, pipelines, model evaluation and cross-validation.
Logistic Regression - linear binary classification.
Random Forest - nonlinear ensemble classification.
SelectKBest / ANOVA F-test - selection of the most informative genes.
Matplotlib and Seaborn - visualization of model performance, ROC curves and confusion matrices.
joblib - saving trained machine learning pipelines for reproducibility and future use.
PyYAML - configuration management.

Results

Dataset Preparation

The final classification dataset consisted of:

- 100 samples
- 77 GBM samples
- 23 non-tumor samples
- 54,613 original probes
- 22,189 genes after probe annotation and aggregation
- 100 selected machine learning features

The data preparation workflow successfully converted the original probe-level microarray data into a gene-level matrix suitable for machine learning.

Model Performance on the Test Set

The models were evaluated on an independent 20-sample test set.

References

1. Polański A., Podstawy bioinformatyki, Wydawnictwo Polsko-Japońskiej Wyższej Szkoły Technik Komputerowych, Warszawa, 2010.
2. Lesk A. M., Wprowadzenie do bioinformatyki, Wydawnictwo Naukowe PWN, Warszawa, 2019.
3. Wang X., Next-Generation Sequencing Data Analysis, 2nd ed., CRC Press, 2024.
4. Korpelainen E., Tuimala J., Somervuo P., Huss M., Wong G., RNA-seq Data Analysis: A Practical Approach, CRC Press, 2015.
5. Azad R. K. (ed.), Transcriptome Data Analysis, Methods in Molecular Biology, vol. 2812, Springer, 2024.
6. A Guide to Applied Machine Learning for Biologists, Springer, 2023.
7. Géron A., Hands-On Machine Learning with Scikit-Learn and PyTorch: Concepts, Tools, and Techniques to Build Intelligent Systems, O'Reilly Media, 2025.
8. Alkhateeb A., Rueda L. (eds.), Machine Learning Methods for Multi-Omics Data Integration, Springer, 2023.
9. Ravarani C., Latysheva N., Deep Learning for Biology, O'Reilly Media, 2025.
10. Chollet F., Watson M., Deep Learning with Python, 3rd ed., Manning Publications, 2025.
11. Chollet F., Kalinowski T., Deep Learning with R, 3rd ed., Manning Publications, 2026.